# General Aptitude

GATE CSE 2027 · complete formula, rule, method & trap sheet

**73** reference boxes · **8** chapters · **79** worked examples (switch on above)

Verbal Aptitude | Numerical Computation | Data Interpretation | Mensuration, Geometry & Probability | Analytical Aptitude | Spatial Aptitude | Problem → Standard Method Playbook | Last-minute one-liners

## 1 Verbal Aptitude

### Grammar essentials GATE tests repeatedly

REFERENCE TABLE

| Area | What to watch for |
|---|---|
| Subject-verb agreement | Singular subject → singular verb, even when other nouns sit in between ("The *list* of items **is** long", not "are") |
| Tenses | Consistency across a sentence; signal words like "since", "for", "yet" hint at the correct tense |
| Prepositions | Fixed idiomatic pairings: "married **to**", "different **from**", "capable **of**" |
| Articles (a/an/the) | "a" before consonant SOUNDS (a university), "an" before vowel SOUNDS (an hour) |

### Vocabulary: idioms & words-in-context

NOTES

GATE rarely tests rote word-lists; it tests whether you can pick the word that fits the **context** and tone of a sentence.

### Reading comprehension strategy

KEY POINT

1. Read the QUESTIONS first (skim), so you know what to look for.
2. Skim the passage once for overall structure/argument, don't over-read details yet.
3. For "which of the following is true/false" questions, go back and verify each option against the specific line in the passage — don't rely on memory or general impression.
4. Watch for **absolute words** in options ("always", "never", "all") — passages rarely support such strong claims, making these options likely wrong.

### Narrative sequencing

NOTES

You're given 4-5 jumbled sentences and must arrange them into a coherent paragraph.

### Sentence completion — a systematic method

NOTES

Single- and double-blank sentence completion is one of the most repeated 1-mark verbal patterns. Do not try the options one by one — read the **logical signal** first.

REFERENCE TABLE

| Signal type | Typical words | What the blank must do |
|---|---|---|
| Contrast | but, although, though, despite, however, yet, while, unlike | Be the **opposite** of the clue |
| Continuation | and, moreover, furthermore, hence, therefore, so, as a result | Carry the **same direction** as the clue |
| Cause and effect | because, since, thus, consequently | Be the logical cause or result of the other clause |
| Definition / example | colon, dash, "that is", "for instance" | Restate the clue in other words |

KEY POINT

**Method:** (1) predict a word BEFORE reading the options; (2) in a two-blank sentence fix the blank you are surer of and eliminate options with it; (3) check the polarity (positive or negative) of both blanks; (4) read the finished sentence back once — it must sound natural, not merely grammatical.

### Critical reasoning — arguments, assumptions, strengthen / weaken

NOTES

An **argument** is premises (evidence) plus a conclusion, joined by an **unstated assumption**. Find the conclusion first (clue words: therefore, hence, thus, so, it follows that), then ask: *what must be true for these premises to lead there?*

REFERENCE TABLE

| Question type | What to look for | The correct option… |
|---|---|---|
| Assumption | The hidden bridge from premise to conclusion | Is **necessary**: negate it and the argument collapses |
| Strengthen | Support for the bridge | Rules out an alternative cause or confirms the link |
| Weaken | A gap in the bridge | Offers an **alternative cause**, or shows the evidence is unrepresentative |
| Inference | What the passage forces | Must be true using ONLY the passage — not merely plausible |

KEY POINT

**Assumption example:** "Profit rose after the firm adopted a four-day week; therefore the four-day week raised profit." The hidden assumption is that *nothing else that affects profit changed at the same time*. Negate it — "something else did change" — and the conclusion no longer follows, which is exactly the test for a necessary assumption.

### Verbal deduction — what MUST follow

NOTES

Deduction questions give a short passage and ask which statement **must be true**. Use only the passage; ignore real-world knowledge.

GATE FAVOURITE

**GATE trap**

"All A are B" does **not** give "All B are A" (the converse). Its valid twin is the **contrapositive**: "if not B then not A". Also watch quantity words — "some" never becomes "all", and "most" never becomes "all".

### Word groups & odd-one-out

NOTES

Find the property shared by three words and use the fourth as the tie-breaker. Test several lenses: **category**, **function**, **degree or intensity**, **part–whole**, **origin**.

REFERENCE TABLE

| Group | Odd one out | Reason |
|---|---|---|
| Brass, bronze, steel, copper | Copper | Copper is an element; the other three are alloys |
| Sprint, dash, dart, stroll | Stroll | The first three are fast movement; a stroll is slow (degree) |
| Lion, tiger, leopard, wolf | Wolf | Big cats versus a canine (biological family) |
| Square, rectangle, rhombus, circle | Circle | The others are quadrilaterals |

# 2 Numerical Computation

## Ratios & Percentages

FORMULA / RULE

**Worked numerical**

**Q:** A shopkeeper marks an item 40% above cost price, then offers a 25% discount on the marked price. Find the net profit/loss percentage.

**A:**

```
Let cost price = 100.
Marked price = 100 × 1.40 = 140
Selling price (after 25% discount) = 140 × 0.75 = 105

Net result = 105 - 100 = 5  → 5% PROFIT
```

Shortcut formula: Net % change = 40 - 25 - (40×25/100) = 40-25-10 = **+5%** (successive percentage change formula: a + b + ab/100, taking discount as negative).

## Powers, Exponents & Logarithms

WORKED EXAMPLE

**Worked numerical**

**Q:** If $\log_{10}2 = 0.301$, find $\log_{10}80$.

**A:**

```
80 = 8 × 10 = 2³ × 10
log(80) = log(2³) + log(10) = 3×log(2) + 1 = 3×0.301 + 1 =
0.903 + 1 = 1.903
```

## Permutations & Combinations

KEY POINT

```
Permutation (ORDER matters):   nPr = n! / (n-r)!
Combination (ORDER doesn't matter): nCr = n! / (r!(n-r)!)
```

## Series & Number Patterns

GATE FAVOURITE

**GATE strategy**

For number series, always compute first differences, then second differences if the first aren't constant. If differences form a clean arithmetic/geometric pattern, you've found the rule. Also always sanity-check with a closed-form formula (like n×(n+1) above) when the pattern is recognizable — it's faster and safer than extrapolating blindly.

## Number system — divisibility, remainders, HCF & LCM

REFERENCE TABLE

| Divisor | Test |
|---|---|
| 2, 5, 10 | Last digit even / 0 or 5 / 0 |
| 3, 9 | Digit sum divisible by 3 / by 9 |
| 4, 8 | Last 2 / last 3 digits divisible by 4 / 8 |
| 6 | Divisible by both 2 and 3 |
| 7 | Drop the last digit d and subtract 2d from the rest; repeat (203 → 20 − 6 = 14 ✔) |
| 11 | (sum of digits in odd places) − (sum in even places) is 0 or a multiple of 11 |

KEY POINT

```
Remainder rules    (a + b) mod m = (a mod m + b mod m) mod m
                   (a × b) mod m = (a mod m × b mod m) mod m
Last-digit cycles  2, 3, 7, 8 repeat every 4 powers;  4, 9
every 2;  0, 1, 5, 6 never change
Fermat             a^(p−1) ≡ 1 (mod p)  for prime p not
dividing a
HCF × LCM = product of the two numbers          (TWO numbers
only)

Greatest number dividing a, b, c and leaving remainders r1,
r2, r3 :  HCF(a−r1, b−r2, c−r3)
Least number leaving the SAME remainder r on division by a,
b, c  :  LCM(a, b, c) + r
Least number that is k SHORT of a multiple of each of a, b,
c      :  LCM(a, b, c) − k
```

## Mixtures & alligation

KEY POINT

```
Alligation (mixing a cheaper item of value c with a dearer
one of value d to get a mean m):

      cheaper : dearer  =  (d − m) : (m − c)

Replacement: a vessel holds V litres of pure liquid; x
litres are drawn off and replaced
by water, n times in all  →  pure liquid left = V × (1 −
x/V)^n
```

## Linear equations & inequalities

KEY POINT

```
Two equations  a1·x + b1·y = c1  and  a2·x + b2·y = c2:
   a1/a2 ≠ b1/b2            → exactly one solution
   a1/a2 = b1/b2 ≠ c1/c2    → no solution (parallel lines)
   a1/a2 = b1/b2 = c1/c2    → infinitely many solutions

Inequalities: multiplying or dividing by a NEGATIVE number
reverses the sign.
   |x − a| < b   ⟺   a − b < x < a + b
   |x − a| > b   ⟺   x < a − b   or   x > a + b
```

## Progressions (AP, GP) and means

KEY POINT

```
AP:  a_n = a + (n−1)d        S_n = (n/2)·[2a + (n−1)d] =
(n/2)·(first + last)
GP:  a_n = a·r^(n−1)         S_n = a(r^n − 1)/(r − 1)
S_∞ = a/(1 − r)   when |r| < 1
For positive numbers: AM ≥ GM ≥ HM; for two numbers  GM² =
AM × HM
```

## Clocks & calendars

KEY POINT

```
Minute hand 6° per minute, hour hand 0.5° per minute →
relative speed 5.5° per minute.
Angle between the hands at H:M  =  |30H − 5.5M|     (use
360° − angle if it exceeds 180°)
In 12 hours the hands coincide 11 times, are opposite 11
times and are at right angles 22 times.

Calendar: ordinary year = 52 weeks + 1 odd day;  leap year =
2 odd days.
Leap year: divisible by 4, but a century year only if
divisible by 400.
Odd days in 100, 200, 300, 400 years = 5, 3, 1, 0.
```

### Time, speed & distance — trains, relative speed, boats

KEY POINT

```
speed = distance / time        km/h → m/s : × 5/18
m/s → km/h : × 18/5
Equal distances at speeds a and b:    average speed = 2ab/(a + b)   (NOT the simple mean)
Same time, speeds a and b:            average speed = (a + b)/2
For a fixed distance, time ∝ 1/speed — a 25 % speed increase cuts the time by 20 %
```

REFERENCE TABLE

| Situation | Distance covered ÷ speed |
|---|---|
| Train (length L) passes a **pole / man** | L ÷ train speed |
| Train passes a **platform / bridge** (length P) | (L + P) ÷ train speed |
| Two trains, **same direction** | ($L_1$ + $L_2$) ÷ **difference** of speeds |
| Two trains, **opposite directions** | ($L_1$ + $L_2$) ÷ **sum** of speeds |

KEY POINT

**Boats and streams.** Boat speed in still water = b, stream = s. **Downstream = b + s**, **upstream = b − s**. From the two speeds: **b = (d + u)/2**, **s = (d − u)/2**. Average speed over a round trip is 2du/(d + u), less than b.

### Time & work — rates, pipes and wages

KEY POINT

```
Work rate = 1/(days to finish alone);  rates ADD for people working together
A in a days, B in b days together:     time = ab/(a + b)
Pick total work = LCM of the days (units) → each person's units/day becomes an integer
Pipes:  inlet = +rate, outlet/leak = −rate        Wages are shared in proportion to WORK DONE
M₁D₁H₁/W₁ = M₂D₂H₂/W₂   (men × days × hours per day ÷ work is constant)
```

### Profit & loss, averages and ages

KEY POINT

```
Profit % = (SP − CP)/CP × 100  (always on COST unless told otherwise)     SP = CP × (1 ± p/100)
Marked price MP, discount d%:   SP = MP × (1 − d/100)
Successive changes multiply: (1 + a)(1 + b)
Average = sum / count.   Replacing a member changes the SUM by (new − old); so  new sum = old sum − old + new
Weighted average = Σ(nᵢ·xᵢ) / Σnᵢ        Multiplying every value by k and adding c: new average = k·avg + c
Ages: let the younger age be x; "after t years" adds t to EVERY age — the DIFFERENCE of ages never changes
```

### Sets & Venn diagrams

KEY POINT

```
|A ∪ B| = |A| + |B| − |A ∩ B|
|A ∪ B ∪ C| = |A| + |B| + |C| − |A∩B| − |B∩C| − |A∩C| + |A∩B∩C|
Fill a Venn diagram from the INSIDE out: start with the triple overlap, then "exactly two" regions
(pairwise overlap − triple), then "only one" regions.
Outside all circles = total − union.
```

### Numerical estimation

KEY POINT

Round to friendly numbers, keep the **order of magnitude**, and compare the answer choices — they are usually far apart. Handy benchmarks: 1/8 = 12.5 %, 1/6 ≈ 16.7 %, 1/3 ≈ 33.3 %, 3/8 = 37.5 %, √2 ≈ 1.414, √3 ≈ 1.732, π ≈ 3.14. Check sums with the **last digit** or the **digit-sum (mod 9)** before computing in full.

```
49.8 × 20.1 / 9.95  ≈  50 × 20 / 10 = 100     (exact: 100.6)
Unit digit of 7^45: the last digits of 7ⁿ cycle 7, 9, 3, 1 (length 4); 45 mod 4 = 1 → unit digit 7
```

## 3 Data Interpretation

### Pie charts — the 360°/100% conversion

KEY POINT

```
Degrees in a pie slice = (category's % of total) × 360° / 100
Percentage from degrees = (slice's degrees / 360°) × 100
```

### Bar graphs & tables — classic multi-step question

WORKED EXAMPLE

**Worked numerical**

**Q:** A bar graph shows a company's profit (in ₹ lakh) over 4 years: 2021: 50, 2022: 60, 2023: 45, 2024: 75. Find: (a) the year-on-year % change from 2022 to 2023, (b) the average annual profit over all 4 years.

**A:**

```
(a) % change (2022→2023) = (45-60)/60 × 100 = -15/60 × 100 = -25%  (a 25% DECREASE)

(b) Average = (50+60+45+75)/4 = 230/4 = ₹57.5 lakh
```

### Common DI traps

GATE FAVOURITE

**GATE angle**

- "% change" ALWAYS divides by the ORIGINAL (older) value, never the newer one.
- When comparing two pie charts with DIFFERENT totals, the same PERCENTAGE can represent very different ACTUAL amounts — always convert to absolute values before comparing across charts.
- Read axis labels and units carefully (lakh vs crore, thousands vs millions) — a classic easy-mark-losing mistake under time pressure.

### Line graphs, multi-series data & caselets

KEY POINT

**Read the scale and the units first** (₹ crore? thousands? percent?). Then:

```
% change = (new − old)/old × 100                (base = the EARLIER year)
Percentage POINTS (difference of two percentages) ≠ percent change:   20 % → 25 % is +5 points but +25 %
Profit % = (revenue − expenditure)/expenditure × 100  (on cost, unless the question says "on revenue")
Ratio questions: compare fractions, don't subtract.
Highest absolute value ≠ highest percentage.
```

COMMON TRAP

**Trap**

If A is 20 % more than B, then B is only 16.67 % less than A — the base changes. Two equal successive percentage steps from 200 to 242 are 10 % each (200 × 1.1² = 242), not 21 %.

# 4 Mensuration, Geometry & Probability

## Mensuration — area & perimeter (2D)

REFERENCE TABLE

| Shape | Area | Perimeter |
|---|---|---|
| Rectangle | l × b | 2(l+b) |
| Triangle | ½ × base × height | sum of 3 sides |
| Circle | $\pi r^2$ | $2\pi r$ (circumference) |
| Trapezium | ½ × (sum of parallel sides) × height | sum of 4 sides |

## Mensuration — volume & surface area (3D)

REFERENCE TABLE

| Solid | Volume | Total Surface Area |
|---|---|---|
| Cube (side a) | $a^3$ | $6a^2$ |
| Cuboid (l,b,h) | l×b×h | 2(lb+bh+hl) |
| Cylinder (r,h) | $\pi r^2 h$ | $2\pi r(r+h)$ |
| Cone (r,h,slant l) | $\frac{1}{3}\pi r^2 h$ | $\pi r(r+l)$ |
| Sphere (r) | $\frac{4}{3}\pi r^3$ | $4\pi r^2$ |

## Geometry basics

MUST REMEMBER

**Pythagoras theorem:** in a right triangle, `(hypotenuse)² = (base)² + (height)²`.
**Similar triangles:** corresponding angles equal, corresponding sides in the same ratio.

## Probability

KEY POINT

```
P(event) = (number of favorable outcomes) / (total possible outcomes)
P(A or B) = P(A) + P(B) - P(A and B)      [for any two events]
P(A and B) = P(A) × P(B)                   [ONLY if A and B are independent]
```

## Elementary Statistics

KEY POINT

```
Mean = (sum of all values) / (number of values)
Median = the middle value when data is sorted (average of 2 middle values if even count)
Mode = the most frequently occurring value
```

# 5 Analytical Aptitude

## Deduction vs Induction

KEY POINT

**Deduction**
Reasoning from general rules to a specific, GUARANTEED conclusion. "All birds have feathers. A sparrow is a bird. ∴ A sparrow has feathers." — if the premises are true, the conclusion MUST be true.

KEY POINT

**Induction**
Reasoning from specific observations to a general (probable, not guaranteed) pattern. "Every sparrow I've seen has feathers. ∴ probably all sparrows have feathers." — likely, but not logically certain.

## Syllogisms — the classic GATE pattern

GATE FAVOURITE

**GATE strategy**
Always test syllogism conclusions by trying to draw a Venn diagram (or a counter-example scenario) where the premises hold true but the conclusion is FALSE. If you can construct even one such scenario, the conclusion is invalid — no matter how "intuitively likely" it feels.

## Analogy

NOTES

Find the relationship between the first pair, then apply the SAME relationship to find the missing term.

## Numerical relations & reasoning

WORKED EXAMPLE

**Worked numerical — coding**
**Q:** If CODE is written as DPEF (each letter shifted +1), how is WORD written?
**A:** Shift each letter forward by 1: W→X, O→P, R→S, D→E → **XPSE**.

WORKED EXAMPLE

**Worked numerical — blood relations**
**Q:** "A is B's father. B is C's sister. D is C's mother." How is A related to D?
**A:** B and C are siblings (B is C's sister, so B is female, C's sibling). A is their father, D is their mother. So **A is D's husband** (both are parents of the same children, B and C).

## Seating arrangements — method

KEY POINT

**Steps:** (1) draw the skeleton — a row or a circle, and note which way people **face**; (2) fix the most constrained person first; (3) place all definite clues before any conditional ones; (4) branch only when forced; (5) re-check every clue against the final arrangement.

**Counting:** n people in a row → n! arrangements; around a circle → (n − 1)! (rotations are the same).

GATE FAVOURITE

**GATE trap**
In a circle, people facing the **centre** have their *left* neighbour clockwise from them; if they face **outward**, left and right swap. A row facing north has left = west and right = east. Always read which way the people face before using "left" or "right".

## Rankings & ordering

KEY POINT

```
Total in a row or class = (rank from one end) + (rank from the other end) − 1
People between two positions = |difference of ranks| − 1
If two people swap places, only the ranks of those two people change.
```

## Truth-teller / liar puzzles

MUST REMEMBER

**Method:** each person is a truth-teller (T) or a liar (L). Assume one person's type, follow what their statement forces on the others, and discard the case at the first contradiction. With 3 people there are only $2^3 = 8$ cases, so a small table is safe. Remember: a liar's statement must be **false as a whole** — the negation of "A and B are both liars" is "at least one of A, B is truthful".

### Direction sense & instruction following

KEY POINT

Track position on a grid: north = +y, east = +x. A **right** turn moves the facing clockwise (N → E → S → W), a **left** turn anticlockwise. Distance from the start = $\sqrt{(\Delta x^2 + \Delta y^2)}$; direction = the direction of the displacement vector. At sunrise your shadow points west.

## 6 Spatial Aptitude

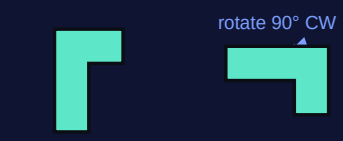
### The five core transformations

REFERENCE TABLE

| Transformation | What changes |
|---|---|
| Translation | Shape slides to a new position — size, orientation, and shape stay identical |
| Rotation | Shape turns around a fixed point by some angle (90°, 180°, etc.) — size stays the same, orientation changes |
| Scaling | Shape grows/shrinks uniformly — orientation stays the same, size changes |
| Mirroring (reflection) | Shape is flipped across a line — becomes a mirror image (left-right or up-down reversed) |
| Assembling/ Grouping | Multiple pieces combine to form a larger recognizable shape (jigsaw-style) |

#### Rotation example

DIAGRAM




Original (L-shape) — After 90° clockwise

NOTES

Notice the L-shape's "foot" that pointed DOWN-RIGHT now points DOWN-LEFT after a 90° clockwise turn — track which specific corner/edge moves where, rather than trying to visualize the whole shape at once.

### Mirror images

COMMON TRAP

**Key rule**

A **mirror image** (reflected left-right, like a vertical mirror) flips horizontal orientation but keeps vertical orientation unchanged. A **water image** (reflected top-bottom, like a reflection in a pond) does the reverse — flips vertical but keeps horizontal unchanged. Letters like **A, H, I, M, O, T, U, V, W, X, Y** look identical in a vertical mirror (they're symmetric left-right); letters like **B, C, D, E, H, I, O, X** look identical in a water/horizontal reflection (symmetric top-bottom).

### Paper folding & cutting

WORKED EXAMPLE

**Worked numerical**

**Q:** A square paper is folded in half twice (first vertically, then horizontally), and a small triangle is cut from the folded corner (the corner where all folds meet). How many holes appear when fully unfolded?

**A:** Each fold doubles the number of layers at that corner. 2 folds → 2×2 = **4 layers** at the folded corner → cutting through all layers at once produces **4 holes** when unfolded, arranged symmetrically (one in each quadrant of the original square).

### Cube/net folding

NOTES

A flattened "net" of 6 squares folds into a cube. The key skill: identify which faces become OPPOSITE each other (opposite faces never share an edge in the net) versus ADJACENT. In a standard cross-shaped net, the four squares in the middle row all become adjacent side faces, while the top and bottom squares fold to become the cube's top and bottom — opposite each other.

### Counting figures & embedded figures

KEY POINT

Count **by size**, never by eye: label the vertices, count all figures of the smallest size, then the next size up, and add. For an embedded (hidden) figure, rotate or reflect the target mentally and check it fits within the larger figure *without altering any line*.

REFERENCE TABLE

| n × n grid | Squares = n(n+1)(2n+1)/6 | Rectangles (all) = [n(n+1)/2]² |
|---|---|---|
| 1 | 1 | 1 |
| 2 | 5 | 9 |
| 3 | 14 | 36 |
| 4 | 30 | 100 |
| 5 | 55 | 225 |

FORMULA / RULE

**Triangles:** if the base of a triangle is divided into n equal parts and every division point is joined to the opposite apex, the number of triangles is n(n+1)/2 (n = 5 gives 15). Non-square rectangles = all rectangles − squares.

### Dice & cubes

KEY POINT

A cube has 6 faces, 8 corners and 12 edges, and **three faces meet at every corner**. So faces seen together in one view are always **adjacent — never opposite**. Ordinary dice have opposite faces summing to 7, but a GATE puzzle does not assume that unless it says so.

**Method:** (1) from each view list the neighbours of every face; (2) a face with four known neighbours has the one remaining face as its opposite; (3) remove numbers already paired and repeat; (4) use the clockwise order around a corner only if adjacency alone does not settle it.

### Figure series, patterns & abstract reasoning

MUST REMEMBER

For a series of figures, **track one feature at a time**: (1) **rotation** per step (angle and direction), (2) **count** change (dots, lines, sides), (3) **shading / colour** toggling, (4) **position** moving around a cycle. Different features usually have **different periods**; the whole figure repeats after the LCM of the periods.

```
Rotation by θ per step repeats the direction when kθ is a
multiple of 360°  → k = 360/gcd(θ, 360)
Direction after k steps = (k × θ) mod 360 :  clockwise from
the start.  Up 0° · Right 90° · Down 180° · Left 270°
Odd-one-out figure: find the property that all but one share
(symmetry, number of sides, shaded area, closed/open).
```

## 7 Problem → Standard Method Playbook

METHOD

**1. Percentage / ratio wording**
Translate the statement into a base quantity → ratio/equation. For successive percentage changes, use multiplication of factors: `new = old × (1±r/100)`. For “more than / less than”, keep the denominator explicit.

METHOD

**2. Remainders / divisibility**
Reduce numbers modulo the divisor. For powers, look for cycles; for divisibility, factor the divisor and apply the relevant tests. Do not assume the remainder distributes over multiplication without taking modulo.

METHOD

**3. Mixture / alligation**
Set quantities around the mean concentration/price. For two components, `quantity of cheaper : quantity of dearer = (dearer−mean) : (mean−cheaper)`.

METHOD

**4. Critical reasoning**
Identify conclusion → premises → hidden assumption. For a necessary assumption, negate the option: if the argument collapses, it is necessary. For strengthen/weaken, ask whether the option changes the causal or logical link rather than merely repeating a premise.

METHOD

**5. Verbal deduction / syllogism**
Translate “all/no/some” into set inclusion/exclusion. Test whether the conclusion follows in *every* model satisfying the premises. “Some A are B” does not imply “some B are all A”.

METHOD

**6. Seating / ranking**
Draw slots. Fix absolute positions first, then relative constraints. For circular seating, remove rotational symmetry by fixing one person. For rankings, convert “from left/right” into a common coordinate before combining.

METHOD

**7. Spatial / cube / dice**
Track adjacency and opposites. Faces seen together are adjacent. For cube rotations, preserve opposite pairs; for counting figures, count by size systematically rather than visually guessing.

METHOD

**8. Speed / time / work**
Convert to one unit system (km/h ↔ m/s: ×5/18). For trains add the lengths and use relative speed; for work questions take total work = LCM of the days and compute units per day. Equal-distance round trips use 2ab/(a+b).

METHOD

**9. Sets, DI and estimation**
Venn: fill from the triple intersection outward. DI: write the formula for the quantity first (profit % is on cost) and separate percent from percentage points. Estimation: round, check the order of magnitude, then eliminate options.

METHOD

**10. Figure series**
Treat each feature (rotation, count, shading, position) as its own sequence and find its period; the combined period is the LCM.

## 9 Last-minute one-liners

LAST-MINUTE · FORMULAS & RULES

- Ranking total = rank from top + rank from bottom − 1; the same person is counted from both ends.
- Alligation ratio = (dearer−mean):(mean−cheaper).
- Squares in an n×n grid = n(n+1)(2n+1)/6; rectangles = [n(n+1)/2]².
- Percentage change = (new-old)/old × 100%; successive percentage changes are multiplicative, not simply additive.
- **For speed, distance, and time:** distance = speed × time; use consistent units before calculation.
- Simple interest = PRT/100; compound interest depends on compounding frequency and period.

LAST-MINUTE · MUST REMEMBER

- **Trains:** pole = L/v; platform = (L+P)/v; two trains = ($L_1$+$L_2$)/relative speed (difference if same direction, sum if opposite).
- **Venn (3 sets):** fill from the triple overlap outward; |A∪B∪C| = Σ singles − Σ pairs + triple.
- In aptitude, translate the wording into a mathematical model before calculating; avoid pattern-matching without checking constraints.
- For ratio questions, multiply all ratio parts by the same positive factor; ratios do not by themselves give absolute quantities.
- In probability, define the sample space carefully; mutually exclusive and independent are different concepts.
- For data interpretation, estimate first when options are widely separated; then calculate only as precisely as necessary.
- Geometry questions should begin by identifying the relevant invariant/standard formula rather than expanding algebra blindly.
- **Critical reasoning:** a weakener offers an alternative cause, a strengthener rules one out, and a necessary assumption collapses the argument when negated.

LAST-MINUTE · COMMON TRAPS

- In seating/cube questions, preserve relative constraints; do not infer adjacency merely from visual placement.
- Average of averages is not generally the average of the group averages unless group sizes are equal or weighted appropriately.
- P(A∪B)=P(A)+P(B)-P(A∩B); do not omit the intersection unless events are mutually exclusive.
- 'All A are B' does not imply 'All B are A'; only the contrapositive (not B → not A) is guaranteed.

LAST-MINUTE · SHORTCUTS

- **Estimation:** round to friendly numbers, check the order of magnitude, eliminate options. Unit digits cycle (7: 7,9,3,1).
- Use approximation, ratio comparison, cancellation, and option elimination whenever the question permits it.

LAST-MINUTE · GATE FAVOURITES

- **Figure series:** track rotation, count, shading and position separately; combined period = LCM of the periods.
- **Number-system questions:** reduce powers modulo the divisor and look for a cycle before calculating large exponents.